

MDSA D208 Predictive Modeling

Task 1

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WGU

Part I: Research Question

A. Describe the purpose of this data analysis by doing the following:

1. Summarize one research question that is relevant to a real-world organizational situation captured in the data set you have selected and that you will answer using multiple linear regression in the initial model.
 - Does the customers monthly charge predict a customers churn? I believe this is an important question to an organization as it costs 10 times more to acquire a new customer than to retain an existing one(Churn Data Consideration and Dictionary 2024).
2. Define the goals of the data analysis.
 - The goal of this analysis is to better understand the relationships, if any, between the MonthlyCharge variable and other variables that may be able to predict a customers churn. The MonthlyCharge variable will be the dependent variable used in the models. We will use multiple independent variables to see if there is a relevant relationship between the dependent variable and the independent variables in order to determine any predictive affects MonthlyCharge may have on a customers churn.

Part II: Method Justification

B. Describe multiple linear regression methods by doing the following:

1. Summarize four assumptions of a multiple linear regression model.

- Multiple linear regression (MLR) model is a statistical technique that uses multiple independent variables to predict the outcome of a dependent variable. Four assumptions of MLR are:
 - Linearity - Assumption there is a linear relationship between dependent and independent variables
 - No Multicollinearity - Assumption that independent variables are not highly correlated with each other
 - Multivariate Normality - Assumption that the residuals are normally distributed
 - No autocorrelation - Assumption that there is little or no autocorrelation in the data. This is also known as independence of independent variables.
2. Describe two benefits of using Python or R in support of various phases of the analysis.
- In this analysis, we will be using Python within a Jupyter notebook. This gives us the ease and flexibility to have code and markdown cells in one easy to follow document. The use of Python allows us to use multiple libraries for MLR that can handle, model and process data. Python combined with Jupyter make it easy for not only implementing and reading code, but also creating the visual representations of the code all in one place making it easier to understand the data that is being worked with.
3. Explain why multiple linear regression is an appropriate technique to use for analyzing the research question summarized in part I.
- Since MonthlyCharge is a numeric variable, multiple linear regression will be used. Multiple linear regression attempts to model a relationship between two or more variables by using an equation to find out which independent variable has the largest impact on the dependent variable. MonthlyCharge will be the dependent variable in this analysis and several independent variables will be used to see if any of the independent variables have an impact on the dependent variable in order to answer the research question from part I.

Part III: Data Preparation

C. Summarize the data preparation process for multiple linear regression analysis by doing the following:

1. Describe your data cleaning goals and the steps used to clean the data to achieve the goals that align with your research question including your annotated code.
- The goal is to have a clean data set that can be used for regression testing mentioned in part B above. The following will be the steps taken in order to clean the data.
 - needed packages will be imported into Python
 - `pd.read_csv` will be used to load data frame
 - `.info()` will be used to profile data frame

- inspect data frame for duplicate values
- `.isnull().sum` will be used to identify possible missing values
- if missing values exist, they will be treated with imputation
- identify outliers
- Below is the code and output for section C1

```
In [67]: # code for c1
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats
import warnings
warnings.filterwarnings('ignore', category=FutureWarning)

# code for data frame
churn = pd.read_csv("C:/users/jjord/Documents/WGU/D208/PA1/churn_clean.csv")

# Profile of Data Frame
churn.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 50 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   CaseOrder             10000 non-null  int64
 1   Customer_id           10000 non-null  object
 2   Interaction            10000 non-null  object
 3   UID                   10000 non-null  object
 4   City                  10000 non-null  object
 5   State                 10000 non-null  object
 6   County                10000 non-null  object
 7   Zip                   10000 non-null  int64
 8   Lat                   10000 non-null  float64
 9   Lng                   10000 non-null  float64
10   Population            10000 non-null  int64
11   Area                  10000 non-null  object
12   TimeZone              10000 non-null  object
13   Job                   10000 non-null  object
14   Children              10000 non-null  int64
15   Age                   10000 non-null  int64
16   Income                10000 non-null  float64
17   Marital               10000 non-null  object
18   Gender                10000 non-null  object
19   Churn                 10000 non-null  object
20   Outage_sec_perweek    10000 non-null  float64
21   Email                 10000 non-null  int64
22   Contacts              10000 non-null  int64
23   Yearly_equip_failure  10000 non-null  int64
24   Techie                10000 non-null  object
25   Contract              10000 non-null  object
26   Port_modem            10000 non-null  object
27   Tablet                10000 non-null  object
28   InternetService       7871 non-null   object
29   Phone                 10000 non-null  object
30   Multiple              10000 non-null  object
31   OnlineSecurity        10000 non-null  object
32   OnlineBackup          10000 non-null  object
33   DeviceProtection      10000 non-null  object
34   TechSupport           10000 non-null  object
35   StreamingTV           10000 non-null  object
36   StreamingMovies       10000 non-null  object
37   PaperlessBilling      10000 non-null  object
38   PaymentMethod         10000 non-null  object
39   Tenure                10000 non-null  float64
40   MonthlyCharge         10000 non-null  float64
41   Bandwidth_GB_Year     10000 non-null  float64
42   Item1                 10000 non-null  int64
43   Item2                 10000 non-null  int64
44   Item3                 10000 non-null  int64
45   Item4                 10000 non-null  int64
46   Item5                 10000 non-null  int64
47   Item6                 10000 non-null  int64
48   Item7                 10000 non-null  int64
49   Item8                 10000 non-null  int64

```

dtypes: float64(7), int64(16), object(27)

memory usage: 3.8+ MB

```
In [68]: # Missing values using isnull  
churn.isnull().sum()
```

```

Out[68]: CaseOrder      0
         Customer_id    0
         Interaction     0
         UID            0
         City           0
         State          0
         County         0
         Zip            0
         Lat            0
         Lng            0
         Population     0
         Area           0
         TimeZone       0
         Job            0
         Children       0
         Age            0
         Income         0
         Marital        0
         Gender         0
         Churn          0
         Outage_sec_perweek 0
         Email          0
         Contacts       0
         Yearly_equip_failure 0
         Techie         0
         Contract       0
         Port_modem     0
         Tablet         0
         InternetService 2129
         Phone          0
         Multiple       0
         OnlineSecurity 0
         OnlineBackup   0
         DeviceProtection 0
         TechSupport    0
         StreamingTV    0
         StreamingMovies 0
         PaperlessBilling 0
         PaymentMethod  0
         Tenure         0
         MonthlyCharge  0
         Bandwidth_GB_Year 0
         Item1          0
         Item2          0
         Item3          0
         Item4          0
         Item5          0
         Item6          0
         Item7          0
         Item8          0
         dtype: int64

```

```

In [69]: # Look at unique values for InternetService
         churn.InternetService.unique()

```

```
Out[69]: array(['Fiber Optic', 'DSL', nan], dtype=object)
```

```
In [70]: # Impute nulls for InternetService
churn["InternetService"].fillna("No Internet Service", inplace=True)
```

```
In [71]: # verify InternetService no longer has nulls
churn["InternetService"].isnull().sum()
```

```
Out[71]: 0
```

Columns Zip, Lat, Lng currently are floats or ints. There are not quantitative so will change datatype to object. These values are not meant for any type of calculation. Since the rest of the data types for categorical are set to object, these will also.

```
In [72]: obj_columns = ["Zip", "Lat", "Lng"]
churn[obj_columns] = churn[obj_columns].astype(object)
```

```
In [73]: # rename item columns
churn.rename(
    columns={
        "Item1": "Timely_response",
        "Item2": "Timely_fixes",
        "Item3": "Timely_replacements",
        "Item4": "Reliability",
        "Item5": "Options",
        "Item6": "Respectful_response",
        "Item7": "Courteous_exchange",
        "Item8": "Active_listening",
    },
    inplace=True,
)
```

```
In [74]: #profile df after clean
churn.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 50 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   CaseOrder             10000 non-null  int64
 1   Customer_id           10000 non-null  object
 2   Interaction            10000 non-null  object
 3   UID                   10000 non-null  object
 4   City                  10000 non-null  object
 5   State                 10000 non-null  object
 6   County                10000 non-null  object
 7   Zip                   10000 non-null  object
 8   Lat                   10000 non-null  object
 9   Lng                   10000 non-null  object
10   Population            10000 non-null  int64
11   Area                  10000 non-null  object
12   TimeZone              10000 non-null  object
13   Job                   10000 non-null  object
14   Children              10000 non-null  int64
15   Age                   10000 non-null  int64
16   Income                10000 non-null  float64
17   Marital               10000 non-null  object
18   Gender                10000 non-null  object
19   Churn                 10000 non-null  object
20   Outage_sec_perweek    10000 non-null  float64
21   Email                 10000 non-null  int64
22   Contacts              10000 non-null  int64
23   Yearly_equip_failure  10000 non-null  int64
24   Techie                10000 non-null  object
25   Contract              10000 non-null  object
26   Port_modem            10000 non-null  object
27   Tablet                10000 non-null  object
28   InternetService       10000 non-null  object
29   Phone                 10000 non-null  object
30   Multiple              10000 non-null  object
31   OnlineSecurity        10000 non-null  object
32   OnlineBackup          10000 non-null  object
33   DeviceProtection      10000 non-null  object
34   TechSupport           10000 non-null  object
35   StreamingTV           10000 non-null  object
36   StreamingMovies       10000 non-null  object
37   PaperlessBilling      10000 non-null  object
38   PaymentMethod         10000 non-null  object
39   Tenure                10000 non-null  float64
40   MonthlyCharge         10000 non-null  float64
41   Bandwidth_GB_Year     10000 non-null  float64
42   Timely_response       10000 non-null  int64
43   Timely_fixes          10000 non-null  int64
44   Timely_replacements   10000 non-null  int64
45   Reliability           10000 non-null  int64
46   Options               10000 non-null  int64
47   Respectful_response   10000 non-null  int64
48   Courteous_exchange    10000 non-null  int64
49   Active_listening      10000 non-null  int64

```

dtypes: float64(5), int64(15), object(30)

memory usage: 3.8+ MB

In [75]: `# code for outliers.
churn.describe()`

Out[75]:

	CaseOrder	Population	Children	Age	Income	Outage_sec_perweek
count	10000.00000	10000.000000	10000.0000	10000.000000	10000.000000	10000.000000
mean	5000.50000	9756.562400	2.0877	53.078400	39806.926771	10.000000
std	2886.89568	14432.698671	2.1472	20.698882	28199.916702	2.000000
min	1.00000	0.000000	0.0000	18.000000	348.670000	0.000000
25%	2500.75000	738.000000	0.0000	35.000000	19224.717500	8.000000
50%	5000.50000	2910.500000	1.0000	53.000000	33170.605000	10.000000
75%	7500.25000	13168.000000	3.0000	71.000000	53246.170000	11.000000
max	10000.00000	111850.000000	10.0000	89.000000	258900.700000	21.000000

C2.

2. Describe the dependent variable and all independent variables using summary statistics that are required to answer the research question, including a screenshot of the summary statistics output for each of these variables.

- Below are the descriptive stats for the dependent variable and all independent variables being used.
 - Dependent Variable - MonthlyCharge
 - Independent Variables - Tenure, Income, Age, Outage_sec_perweek, Yearly_equip_failure, Bandwidth_GB_Year, InternetService, Phone, Gender

In [76]: `#c2 code descriptive stats for dependent variable MonthlyCharge
churn['MonthlyCharge'].describe()`

Out[76]:

count	10000.000000
mean	172.624816
std	42.943094
min	79.978860
25%	139.979239
50%	167.484700
75%	200.734725
max	290.160419

Name: MonthlyCharge, dtype: float64

Descriptive stats for independent Variables - Tenure, Income, Age, Outage_sec_perweek, Yearly_equip_failure, Bandwidth_GB_Year, InternetService, Phone, Gender

```
In [77]: # descriptive stats for independent Variables  
churn['Tenure'].describe()
```

```
Out[77]: count      10000.000000  
mean         34.526188  
std          26.443063  
min           1.000259  
25%           7.917694  
50%          35.430507  
75%          61.479795  
max          71.999280  
Name: Tenure, dtype: float64
```

```
In [78]: # descriptive stats for independent Variables  
churn['Income'].describe()
```

```
Out[78]: count      10000.000000  
mean      39806.926771  
std      28199.916702  
min       348.670000  
25%     19224.717500  
50%     33170.605000  
75%     53246.170000  
max     258900.700000  
Name: Income, dtype: float64
```

```
In [79]: # descriptive stats for independent Variables  
churn['Age'].describe()
```

```
Out[79]: count      10000.000000  
mean         53.078400  
std          20.698882  
min          18.000000  
25%          35.000000  
50%          53.000000  
75%          71.000000  
max          89.000000  
Name: Age, dtype: float64
```

```
In [80]: # descriptive stats for independent Variables  
churn['Outage_sec_perweek'].describe()
```

```
Out[80]: count      10000.000000  
mean         10.001848  
std           2.976019  
min           0.099747  
25%           8.018214  
50%          10.018560  
75%          11.969485  
max          21.207230  
Name: Outage_sec_perweek, dtype: float64
```

```
In [81]: # descriptive stats for independent Variables  
churn['Yearly_equip_failure'].describe()
```

```
Out[81]: count    10000.000000
         mean      0.398000
         std       0.635953
         min       0.000000
         25%       0.000000
         50%       0.000000
         75%       1.000000
         max       6.000000
         Name: Yearly_equip_failure, dtype: float64
```

```
In [82]: # descriptive stats for independent Variables
         churn['Bandwidth_GB_Year'].describe()
```

```
Out[82]: count    10000.000000
         mean     3392.341550
         std      2185.294852
         min      155.506715
         25%     1236.470827
         50%     3279.536903
         75%     5586.141370
         max     7158.981530
         Name: Bandwidth_GB_Year, dtype: float64
```

```
In [83]: # descriptive stats for independent Variables
         print(churn['InternetService'].describe())
         print()
         print(churn['InternetService'].value_counts())
```

```
count    10000
unique      3
top      Fiber Optic
freq      4408
Name: InternetService, dtype: object
```

```
InternetService
Fiber Optic      4408
DSL              3463
No Internet Service  2129
Name: count, dtype: int64
```

```
In [84]: # descriptive stats for independent Variables
         print(churn['Phone'].describe())
         print()
         print(churn['Phone'].value_counts())
```

```
count    10000
unique      2
top        Yes
freq      9067
Name: Phone, dtype: object
```

```
Phone
Yes    9067
No     933
Name: count, dtype: int64
```

```
In [85]: # descriptive stats for independent Variables
print(churn['Gender'].describe())
print()
print(churn['Gender'].value_counts())
```

```
count      10000
unique         3
top      Female
freq       5025
Name: Gender, dtype: object
```

```
Gender
Female      5025
Male       4744
Nonbinary    231
Name: count, dtype: int64
```

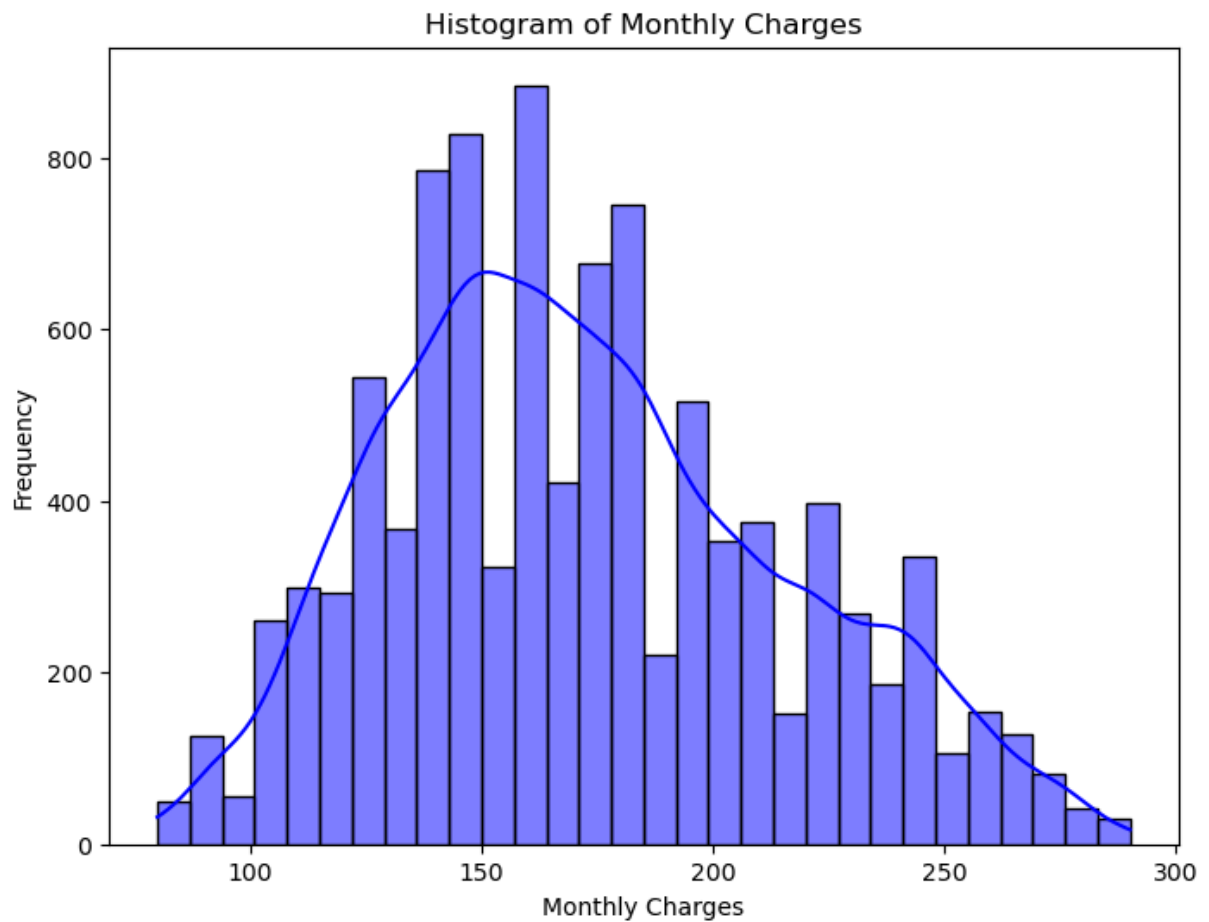
C3.

3. Generate univariate and bivariate visualizations of the distributions of the dependent and independent variables, including the dependent variable in your bivariate visualizations.

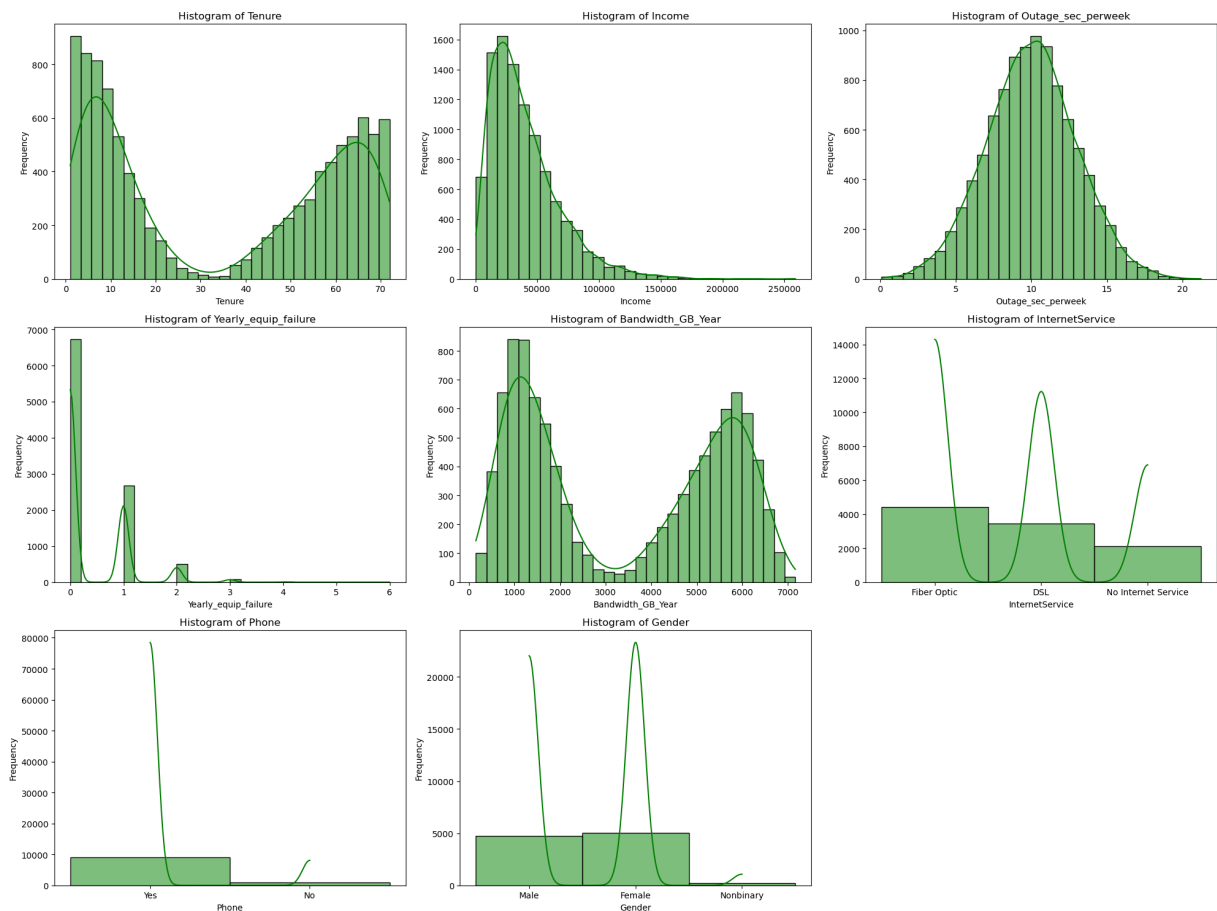
- Below is the code and output for the univariate and bivariate visualizations for the dependent and independent variables.

```
In [86]: #c3 code visualizations for univariate
# MonthlyCharge Histogram

plt.figure(figsize=(8, 6))
sns.histplot(churn['MonthlyCharge'], kde=True, bins=30, color='blue')
plt.title('Histogram of Monthly Charges')
plt.xlabel('Monthly Charges')
plt.ylabel('Frequency')
plt.show()
```

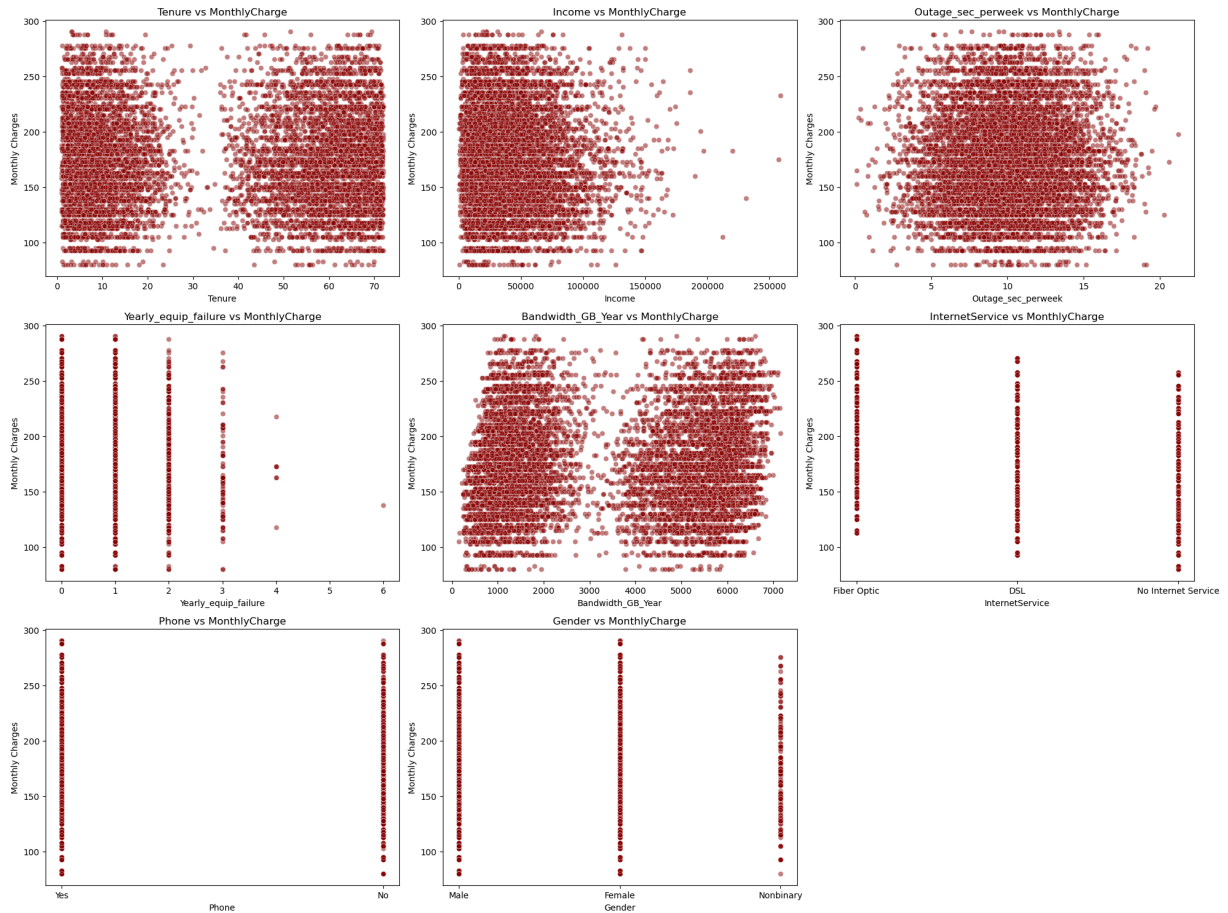


```
In [87]: #c3 cont. code visualizations for univariate
#Independent variable histograms
plt.figure(figsize=(20, 15))
for x, col in enumerate(['Tenure', 'Income', 'Outage_sec_perweek', 'Yearly equip_fa
plt.subplot(3, 3, x)
sns.histplot(churn[col], kde=True, bins=30, color='green')
plt.title(f'Histogram of {col}')
plt.xlabel(col)
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



In [88]: *#code for bivariate visualizations for dependent variable(MonthlyCharge) against ea*

```
plt.figure(figsize=(20, 15))
for x, col in enumerate(['Tenure', 'Income', 'Outage_sec_perweek', 'Yearly equip_fa
plt.subplot(3, 3, x)
sns.scatterplot(data=churn, x=col, y='MonthlyCharge', color='darkred', alpha=0.
plt.title(f'{col} vs MonthlyCharge')
plt.xlabel(col)
plt.ylabel('Monthly Charges')
plt.tight_layout()
plt.show()
```



4. Describe your data transformation goals that align with your research question and the steps used to transform the data to achieve the goals, including the annotated code.

- The goals for data transformation will be to create new columns of the categorical variables that will be transformed into numeric variables. The numeric variables are stored in new columns, preserving the originals. Dummy variables will not need to be encoded since the model will use the transformed columns for the analysis.

5. Provide the prepared data set as a CSV file.

Below is the code for both C4 & C5

```
In [89]: #c4 & c5 code
#create new columns
churn['InternetService_num']=churn['InternetService']
churn['Phone_num']=churn['Phone']
churn['Gender_num']=churn['Gender']

In [90]: #dictionary creation
dict_InternetService = {"Fiber Optic":0, "DSL":1, "No Internet Service":2}

dict_phone = {"No":0, "Yes":1}
```

```
dict_gender = {"Male":0, "Female":1, "Nonbinary":2}
```

```
In [91]: #replace values from dictionaries above
churn['InternetService_num'].replace(dict_InternetService, inplace=True)
churn['Phone_num'].replace(dict_phone, inplace=True)
churn['Gender_num'].replace(dict_gender, inplace=True)
```

```
In [92]: #verify new columns
churn.columns
```

```
Out[92]: Index(['CaseOrder', 'Customer_id', 'Interaction', 'UID', 'City', 'State',
               'County', 'Zip', 'Lat', 'Lng', 'Population', 'Area', 'TimeZone', 'Job',
               'Children', 'Age', 'Income', 'Marital', 'Gender', 'Churn',
               'Outage_sec_perweek', 'Email', 'Contacts', 'Yearly_equip_failure',
               'Techie', 'Contract', 'Port_modem', 'Tablet', 'InternetService',
               'Phone', 'Multiple', 'OnlineSecurity', 'OnlineBackup',
               'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies',
               'PaperlessBilling', 'PaymentMethod', 'Tenure', 'MonthlyCharge',
               'Bandwidth_GB_Year', 'Timely_response', 'Timely_fixes',
               'Timely_replacements', 'Reliability', 'Options', 'Respectful_response',
               'Courteous_exchange', 'Active_listening', 'InternetService_num',
               'Phone_num', 'Gender_num'],
              dtype='object')
```

```
In [93]: #c5 prepped data set
churn.to_csv('prep_churn.csv', index=False)
```

Part IV: Model Comparison and Analysis

D. Compare an initial and a reduced linear regression model by doing the following:

1. Construct an initial multiple linear regression model from all independent variables that were identified in part C2.

```
In [94]: #d1 code
#create new dataframe for model
churn_model = churn[['MonthlyCharge', 'Tenure', 'Income', 'Age', 'Outage_sec_perweek']]
churn_model
```

Out[94]:

	MonthlyCharge	Tenure	Income	Age	Outage_sec_perweek	Yearly_equip_failure
0	172.455519	6.795513	28561.99	68	7.978323	1
1	242.632554	1.156681	21704.77	27	11.699080	1
2	159.947583	15.754144	9609.57	50	10.752800	1
3	119.956840	17.087227	18925.23	48	14.913540	0
4	149.948316	1.670972	40074.19	83	8.147417	1
...	...	...	...	...	...	...
9995	159.979400	68.197130	55723.74	23	9.415935	0
9996	207.481100	61.040370	34129.34	48	6.740547	0
9997	169.974100	47.416890	45983.43	48	6.590911	0
9998	252.624000	71.095600	16667.58	39	12.071910	0
9999	217.484000	63.350860	9020.92	28	11.754720	0

10000 rows × 10 columns



In [95]:

```
#import statsmodel
import statsmodels.api as sm
```

In [96]:

```
# D1 initial model
#variables
x = churn_model[['Tenure', 'Income', 'Age', 'Outage_sec_perweek', 'Yearly_equip_fail
y = churn_model['MonthlyCharge']

#add constant to independent variables
x = sm.add_constant(x)

#fit multiple linear regression model
model = sm.OLS(y,x).fit()

#print model summary
print(model.summary())
```

OLS Regression Results

=====						
Dep. Variable:	MonthlyCharge	R-squared:	0.415			
Model:	OLS	Adj. R-squared:	0.414			
Method:	Least Squares	F-statistic:	786.0			
Date:	Tue, 04 Mar 2025	Prob (F-statistic):	0.00			
Time:	19:16:11	Log-Likelihood:	-49111.			
No. Observations:	10000	AIC:	9.824e+04			
Df Residuals:	9990	BIC:	9.831e+04			
Df Model:	9					
Covariance Type:	nonrobust					
=====						
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	118.2819	2.136	55.367	0.000	114.094	122.469
Tenure	-7.3604	0.101	-73.186	0.000	-7.558	-7.163
Income	-1.067e-05	1.17e-05	-0.915	0.360	-3.35e-05	1.22e-05
Age	0.3243	0.016	19.771	0.000	0.292	0.356
Outage_sec_perweek	0.2464	0.110	2.231	0.026	0.030	0.463
Yearly equip_failure	-0.4573	0.517	-0.884	0.376	-1.471	0.556
Bandwidth_GB_Year	0.0897	0.001	73.682	0.000	0.087	0.092
InternetService_num	-21.8706	0.429	-51.012	0.000	-22.711	-21.030
Phone_num	-1.1123	1.131	-0.984	0.325	-3.328	1.104
Gender_num	5.6017	0.613	9.144	0.000	4.401	6.803
=====						
Omnibus:	94.752	Durbin-Watson:	1.975			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	68.380			
Skew:	0.095	Prob(JB):	1.42e-15			
Kurtosis:	2.642	Cond. No.	3.31e+05			
=====						

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.31e+05. This might indicate that there are strong multicollinearity or other numerical problems.

D2

2. Justify a statistically based feature selection procedure or a model evaluation metric to reduce the initial model in a way that aligns with the research question.

- Backward elimination will be used to remove variables that are insignificant based on their p-values. P-values with a significance level higher than .05 will be removed. By doing this, it will remove variables that do not have a significant impact on the dependent variable.
3. Provide a reduced linear regression model that follows the feature selection or model evaluation process in part D2, including a screenshot of the output for each model.
- Code for D3 is below

```
In [97]: #code for d2 & d3
x = sm.add_constant(x)

#initial model
model = sm.OLS(y,x).fit()

#backward elimination
while True:
    p_values = model.pvalues[1:]
    max_p_value = p_values.max()
    if max_p_value > 0.05:
        max_p_index = p_values.idxmax()
        x = x.drop(columns=[max_p_index])
        model = sm.OLS(y, x).fit()
    else:
        break
#print model summary
print(model.summary())
```

OLS Regression Results

Dep. Variable:	MonthlyCharge	R-squared:	0.414
Model:	OLS	Adj. R-squared:	0.414
Method:	Least Squares	F-statistic:	1179.
Date:	Tue, 04 Mar 2025	Prob (F-statistic):	0.00
Time:	19:17:19	Log-Likelihood:	-49112.
No. Observations:	10000	AIC:	9.824e+04
Df Residuals:	9993	BIC:	9.829e+04
Df Model:	6		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.9

const	116.6806	1.801	64.788	0.000	113.150	120.
Tenure	-7.3608	0.101	-73.206	0.000	-7.558	-7.
Age	0.3241	0.016	19.760	0.000	0.292	0.
Outage_sec_perweek	0.2483	0.110	2.247	0.025	0.032	0.
Bandwidth_GB_Year	0.0897	0.001	73.700	0.000	0.087	0.
InternetService_num	-21.8798	0.429	-51.041	0.000	-22.720	-21.
Gender_num	5.5858	0.612	9.121	0.000	4.385	6.

Omnibus:	95.188	Durbin-Watson:	1.975
Prob(Omnibus):	0.000	Jarque-Bera (JB):	68.636
Skew:	0.095	Prob(JB):	1.25e-15
Kurtosis:	2.641	Cond. No.	2.22e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.22e+04. This might indicate that there are strong multicollinearity or other numerical problems.

```
In [98]: #code for d3
x = sm.add_constant(x)

#fit reduced model
reduced_model = sm.OLS(y,x[['const','Tenure','Age','Outage_sec_perweek','Bandwidth_

#print reduced model summary
print(reduced_model.summary())
```

OLS Regression Results

```

=====
Dep. Variable:          MonthlyCharge    R-squared:                 0.414
Model:                  OLS              Adj. R-squared:          0.414
Method:                 Least Squares    F-statistic:              1179.
Date:                   Tue, 04 Mar 2025  Prob (F-statistic):      0.00
Time:                   19:17:40          Log-Likelihood:           -49112.
No. Observations:      10000             AIC:                     9.824e+04
Df Residuals:          9993              BIC:                     9.829e+04
Df Model:               6
Covariance Type:       nonrobust
=====
===

```

	coef	std err	t	P> t	[0.025	0.9
75]						

const	116.6806	1.801	64.788	0.000	113.150	120.
211						
Tenure	-7.3608	0.101	-73.206	0.000	-7.558	-7.
164						
Age	0.3241	0.016	19.760	0.000	0.292	0.
356						
Outage_sec_perweek	0.2483	0.110	2.247	0.025	0.032	0.
465						
Bandwidth_GB_Year	0.0897	0.001	73.700	0.000	0.087	0.
092						
InternetService_num	-21.8798	0.429	-51.041	0.000	-22.720	-21.
040						
Gender_num	5.5858	0.612	9.121	0.000	4.385	6.
786						
=====						
Omnibus:	95.188	Durbin-Watson:	1.975			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	68.636			
Skew:	0.095	Prob(JB):	1.25e-15			
Kurtosis:	2.641	Cond. No.	2.22e+04			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.22e+04. This might indicate that there are strong multicollinearity or other numerical problems.

E. Analyze the data set using your reduced linear regression model by doing the following:

1. Explain your data analysis process by comparing the initial multiple linear regression model and reduced linear regression model, including the following element:

- Backward elimination was used to remove variables that are insignificant based on their p-values. The model evaluation metric used was R squared. In both models, the R squared valued was almost the same. However, the F statistic value improved by a good amount after

reducing the independent variables. P-values with a significance level higher than .05 were removed.

- Original Model R squared value = .415
- Reduced Model R squared value = .414
- Original Model F statistic value = 786
- Reduced Model F statistic value = 1179

E2

2. Provide the output and all calculations of the analysis you performed, including the following elements for your reduced linear regression model:

- a residual plot
- the model's residual standard error

3. Provide an executable error-free copy of the code used to support the implementation of the linear regression models using a Python or R file.

- below is the code for E2 & E3

```
In [99]: #code for e2 & e3
x = sm.add_constant(x)

#fit reduced model
reduced_model = sm.OLS(y,x[['const', 'Tenure', 'Age', 'Outage_sec_perweek', 'Bandwidth_

#residual and prediction calculations
predictions = reduced_model.predict()
residuals = reduced_model.resid

#summary stats calculation for residuals
residual_summary = {
    'Mean' : np.mean(residuals),
    'Median' : np.median(residuals),
    'Standard Deviation' : np.std(residuals),
    'Minimum' : np.min(residuals),
    'Maximum' : np.max(residuals)
}

#print summary stats
print("Summary of Residuals")
for stat, value in residual_summary.items():
    print(f"{stat}: {value}")

#plots for residuals
```

```
plt.figure(figsize=(8, 6))
plt.scatter(predictions, residuals, color='blue', alpha=0.5)
plt.axhline(y=0, color='red', linestyle='--')
plt.title('Residual Plot for Reduced Linear Regression Model')
plt.xlabel('Predicted Values')
plt.ylabel('Residuals')
plt.show()

#RSE
rse = np.sqrt(np.mean(residuals**2))

# Print the residual standard error
print(f"Residual Standard Error: {rse}")
```

Summary of Residuals

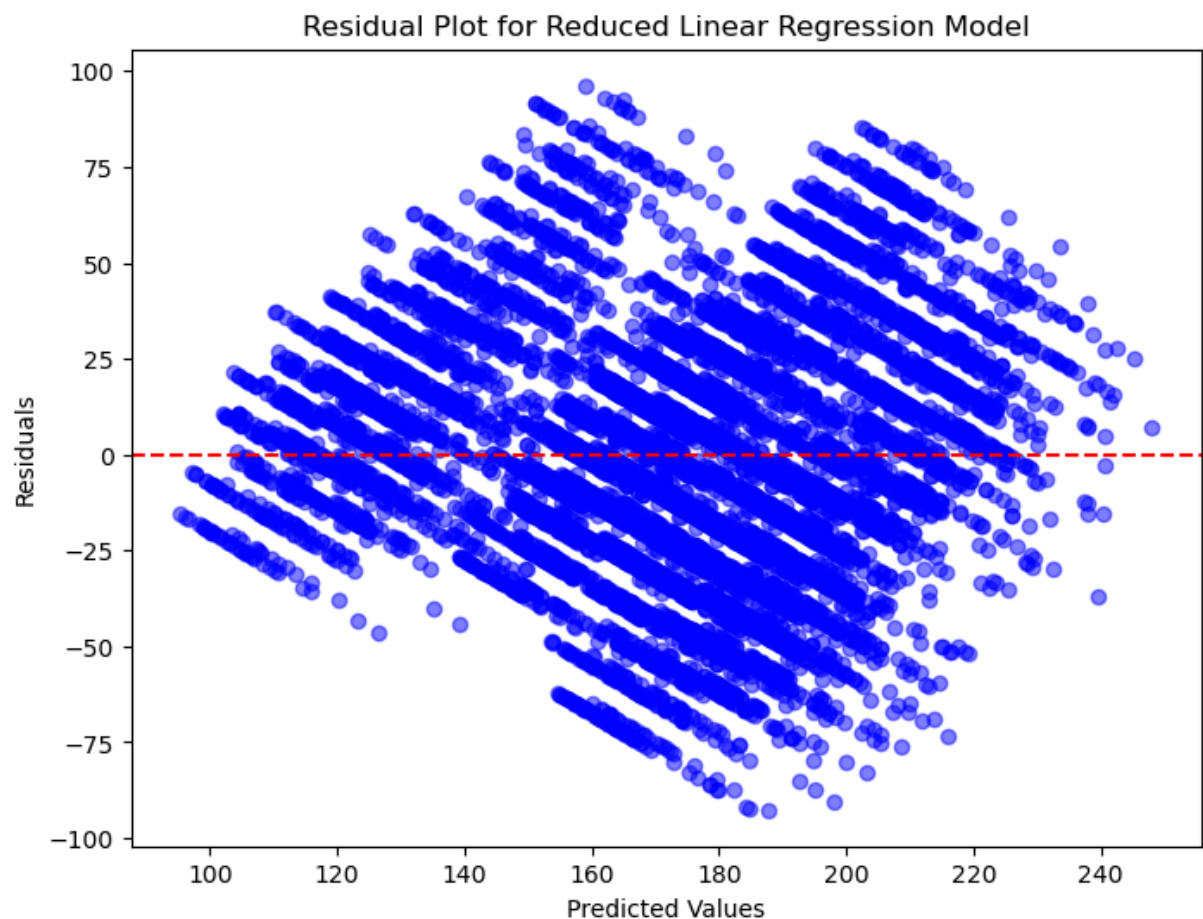
Mean: 1.772641553543508e-11

Median: -0.8775084579627475

Standard Deviation: 32.8601171762473

Minimum: -92.6878695877903

Maximum: 96.24915924618924



Residual Standard Error: 32.8601171762473

Part V: Data Summary and Implications

F. Summarize your findings and assumptions by doing the following:

1. Discuss the results of your data analysis, including the following elements:

- a regression equation for the reduced model - $Y = 116.6806 + -7.3608(\text{Tenure}) + .3241(\text{Age}) + .2483(\text{Outage_sec_perweek}) + .0897(\text{Bandwidth_BG_Year}) + -21.8798(\text{InternetService_num}) + 5.5858(\text{Gender_num})$
 - an interpretation of the coefficients of the reduced model - Tenure decreased monthlycharge by 7.3608 - Age increase montlycharge by .3241 - Outage_sec_perweek increased monthlycharge by .2483 - Bandwidth_GB_Year increased monthlycharge by .0897 - InternetService_num decreased montlycharge by 21.8798 - Gender_num increased montlycharge by 5.5858
 - the statistical and practical significance of the reduced model
 - With the R squared value being .414, it suggests that model only explains 41.4% of the variance. The lower R squared shows there is a variance in the dependent variable not explained by the independent variables used. The statistical significance of the model is mediocre at best.
 - There really isn't much of a practical significance of the reduced model. The relationship between the vairables suggest too low of a variance to be acceptable based upon the R squared value.
 - the limitations of the data analysis
 - There are several limitations with this analysis. First, the standard error can be large. Second, regression models are limited by correlation does not equal causation. The model indicates varaibles that are most correlated with MonthlyCharge but it cannot be said that just because a customer uses more Bandwidth_GB_year that they pay more per month. Lastly, with the R squared valued being low, it would also be considered a limitation.
- ### 2. Recommend a course of action based on your results.
- I would recommend that the data be furhter explored with different variables to see if a customers churn can be predicted by their monthly charge. Based on the variables used, there is not enough variance here to accuratly predict churn. After further analysis has been done, the company can then make an informed decision on the research question.

Part VI: Demonstration

G. Provide a Panopto video recording that includes the presenter and a vocalized demonstration of the functionality of the code used for the analysis of the programming environment, including the following elements:

- an identification of the version of the programming environment
- a comparison of the initial multiple linear regression model you used and the reduced linear regression model you used in your analysis
- an interpretation of the coefficients of the reduced model
- <https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=6df0b42d-bfcf-4c90-a60d-b297000224f3>

H. List the web sources used to acquire data or segments of third-party code to support the application. Ensure the web sources are reliable.

I. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

WGU. (n.d.). Churn Data Consideration and Dictionary. Data Files and Associated Dictionary Files.
<https://access.wgu.edu/ASP3/aap/content/d9rkejv84kd9rk30fi2l.zip>